

YAKIMA, WA, USA

WMO: 727810

Lat: 46.568N

Lon: 120.543W

Elev: 1064

StdP: 14.14

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24243

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.2	14.4	1.3	6.1	14.5	6.7	8.1	17.6	23.2	44.8	18.8	45.8	4.4	270	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	97.4	66.0	94.1	65.3	90.7	63.9	68.3	91.8	66.5	89.8	65.1	87.5	5.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	79.9	75.9	57.5	73.4	74.3	55.5	68.3	74.1	33.0	92.1	31.6	89.9	30.5	87.4	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	18.5	15.9	1.5	102.3	8.1	2.7	-4.4	104.2	-9.1	105.8	-13.7	107.3	-19.5	109.3
			1.1	72.7	7.5	4.9	-4.2	76.2	-8.6	79.1	-12.8	81.8	-18.2	85.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	31.9	36.5	43.6	50.1	58.9	65.3	73.0	71.0	62.2	49.8	38.5	30.5	(6)
	DBStd	16.02	7.60	7.33	6.32	6.40	7.44	7.04	6.49	6.07	6.54	6.80	7.83	7.60	(7)
	HDD50	2285	560	379	213	76	11	0	0	0	3	85	353	605	(8)
	HDD65	5730	1025	797	665	448	219	80	11	17	129	473	796	1070	(9)
	CDD50	2653	0	1	13	79	286	458	714	650	367	78	7	0	(10)
	CDD65	624	0	0	0	1	29	88	260	202	43	1	0	0	(11)
	CDH74	10510	0	0	1	69	619	1492	4075	3208	1006	40	0	0	(12)
	CDH80	5017	0	0	0	13	213	629	2176	1604	375	7	0	0	(13)
(14) Wind	WSAvg	5.8	3.9	5.2	6.5	7.1	7.0	7.3	6.8	6.4	5.9	5.4	4.6	3.7	(14)
(15) Precipitation	PrecAvg	7.8	1.1	0.8	0.7	0.5	0.6	0.5	0.2	0.3	0.3	0.6	0.9	1.4	(15)
	PrecMax	13.2	3.7	2.4	2.0	1.6	2.6	2.5	0.7	2.1	2.1	2.4	2.8	4.2	(16)
	PrecMin	4.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
	PrecStd	1.9	0.7	0.6	0.5	0.4	0.6	0.5	0.2	0.5	0.4	0.5	0.6	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	61.3	70.3	81.8	91.6	98.4	102.1	100.2	94.0	79.1	65.2	54.5	(19)
		MCWB	46.2	49.4	53.1	58.1	62.1	66.5	67.6	67.3	65.3	58.7	55.0	46.8	(20)
	2%	DB	50.3	55.8	65.0	74.7	86.0	91.8	98.5	96.3	88.3	72.3	59.3	49.3	(21)
		MCWB	42.8	45.5	50.1	55.8	60.9	64.3	66.3	65.9	63.1	55.3	49.4	42.7	(22)
	5%	DB	46.0	51.8	60.8	69.6	81.0	87.0	95.2	92.9	84.5	68.0	55.3	44.9	(23)
		MCWB	40.5	42.8	47.9	52.7	58.7	61.8	65.4	65.0	61.3	53.3	46.8	39.7	(24)
	10%	DB	42.1	48.5	57.5	65.2	76.1	82.7	91.7	89.2	79.9	64.3	51.6	41.1	(25)
		MCWB	38.1	40.9	45.9	50.2	56.7	60.1	64.3	63.6	59.4	51.3	44.5	37.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	50.5	54.4	60.2	64.3	68.8	71.8	70.4	67.0	60.0	55.4	47.9	(27)
		MCDB	53.9	60.4	67.3	77.2	85.8	92.9	94.5	92.7	88.4	74.9	63.7	53.7	(28)
	2%	WB	43.4	46.3	51.4	56.8	62.1	65.4	68.8	67.9	64.7	57.1	50.8	43.6	(29)
		MCDB	49.1	54.0	62.9	72.7	82.2	88.5	92.2	91.5	85.7	69.4	57.7	48.5	(30)
	5%	WB	41.1	43.9	49.1	53.7	60.1	63.1	67.1	66.2	62.6	54.7	48.1	40.4	(31)
		MCDB	45.6	50.6	59.5	67.5	77.9	84.7	90.9	88.9	81.1	65.6	54.0	44.2	(32)
	10%	WB	38.5	41.7	46.8	51.1	57.9	61.1	65.5	64.6	60.6	52.5	45.4	37.6	(33)
		MCDB	41.5	47.6	56.2	63.6	73.8	80.5	88.5	86.5	77.2	62.4	50.7	40.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	19.0	24.1	26.7	27.9	28.9	32.8	33.0	32.2	26.9	20.1	13.6	(35)
		MCDBR	20.6	24.5	29.4	33.2	35.2	34.7	36.9	37.2	37.9	31.3	23.8	20.0	(36)
	5% WB	MCWBR	14.5	15.5	16.9	17.1	14.7	13.4	13.1	14.1	16.5	17.1	15.6	14.4	(37)
		MCWBR	19.2	22.3	27.4	31.1	31.4	32.7	34.5	34.8	35.2	28.4	22.2	18.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.293	0.299	0.305	0.325	0.336	0.336	0.339	0.337	0.330	0.319	0.308	0.288	(40)
		taud	2.485	2.514	2.541	2.473	2.466	2.486	2.488	2.489	2.496	2.521	2.496	2.494	(41)
	Ebn at Noon		258	281	295	296	295	294	292	289	281	269	249	246	(42)
		Edh at Noon	21	24	28	32	34	33	33	31	29	24	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		384	694	1065	1560	1899	2169	2386	2014	1481	879	466	309	(44)
	RadStd		48	82	84	109	132	123	115	113	93	94	50	45	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+1.03	N/A	N/A	+2.11	N/A	N/A	N/A	N/A	+279	+146	
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+67	(47)

Nomenclature: See separate page